

a



A consensus database of the accessible human surfaceome.

Roughly 10–20% of human proteins reach the cell surface, yet over 65% of approved drugs target these molecules — accessibility on the extracellular face of the plasma membrane lets a drug act without crossing into the cell. Five public databases (UniProt, GO Cellular Component, the Human Protein Atlas, the Cell Surface Protein Atlas, and SURFY) each catalog the surfaceome, but with different definitions and methods, so their calls disagree more often than they agree.

This atlas reconciles them. An LLM triage agent scores every protein-coding human gene against the union of the five sources; a deep-dive agent then assembles the evidence behind each strong candidate — surface-localization methods, antibody validation, isoform topology, accessibility caveats — into per-gene records. Open accession, evidence-cited, agent-readable.

CURATED LISTS ⓘ Canonical 1,783 ⓘ Likely 1,252 ⓘ Cell-state induced 454 ⓘ Cell-type restricted 852 ⓘ Low-lit + UniProt 538 ⓘ

Filter by symbol, UniProt, name, or synonym... All 19,324 Deep dive 5332 More filters TSV ↓ 19,324

19,324 genes TSV is verdicts + reason codes only. Free-text reasoning per run is on GET /v1/triage/{SYMBOL}; the full deep-dive SurfaceomeRecord is on GET /v1/genes/{SYMBOL}.

SYMBOL	DB VOTES	UNIPROT	GO	SURFY	CSPA	HPA	TRIAGE AGENT	TRIAGE REASON	DEEP-DIVE REASON	DEEP DIVE	CONF	EVIDENCE	STATE DEP.
ABCA1 ATP binding cassette subfamily A member 1	3	●	○	●	○	●	YES	Multi-pass	Multi-pass	HIGH	HIGH	MULTI	MODERATE

b

← Jump to gene... JSON ⌵ Markdown (full) ⌵

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SRC

HGNC HGNC:11283 UNIPROT P12931 NCBI GENE 6714 ENSEMBL ENSG00000197122 Submit feedback ⌵

Likely ⓘ Cell-state induced ⓘ

SOURCES 2/5 ● UniProt ● GO ● SURFY ● CSPA ● HPA

SURFACE LIKELIHOOD ⓘ
Moderate Architecture Other ⓘ Family Enzyme ⓘ

SRC is state-dependently surface-accessible in cancer cells — a canonical inner-leaflet kinase with a cancer-state outer-leaflet form generated by autophagolysosomal exocytosis (ALE). Direct single-method support: functional antibody-mediated tumor killing in vitro and in mouse xenograft models targeting extracellular membrane-associated Src (eSrc) confirms outer-leaflet display in cancer cells (1, 2). Surface presence is strictly state-gated, requiring oncogenic transformation; under normal conditions SRC is exclusively inner-leaflet-anchored via N-myristoylation with no extracellular domain (3, 4, 17). Moderate conformational epitope masking is the principal binder-engineering caveat — the SH2/SH3 intramolecular closed-state switch may occlude epitopes on the inverted eSrc form (3); the absence of a shed or secreted free-soluble form rules out a decoy concern.

EXPERIMENTAL SURFACE EVIDENCE ⓘ
Direct, single method
37 entries

Reasoning ⌵

CONFIDENCE ⓘ
Low
26 primary · 11 secondary

Reasoning ⌵

STATE DEPENDENCE ⓘ
High
Reasoning ⌵

Canonical P12931	Isoform 2 P12931-2	Isoform 3 P12931-3	Mouse ortholog P05480
Cyno ortholog A0A7N9CC30		Experimental 8JN8 · A	



AlphaFold model
WHOLE PLDDT 83.4 ⓘ
Disordered 19% ⓘ AFDB P12931 · v6 ⌵
● Globular

